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Studierichting: Informatica
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$$x_n = (-1)^n \left(1 + \frac{1}{n}\right)$$

1. n even: $x_{2n} = 1 + \frac{1}{2n}$

$$\lim_{n \rightarrow +\infty} x_{2n} = 1$$

2. n oneven: $x_{2n-1} = -\left(1 + \frac{1}{2n-1}\right)$

$$\lim_{n \rightarrow +\infty} x_{2n-1} = -1$$

x_n is afwisselend en nadert -1 en 1

$$\Rightarrow \lim_{n \rightarrow +\infty} x_n \text{ bestaat niet}$$

$$\Rightarrow \text{adh}_{n \rightarrow +\infty} (x_n) = \{-1, 1\}$$

$$\Rightarrow \liminf_{n \rightarrow +\infty} x_n = -1$$

$$\Rightarrow \limsup_{n \rightarrow +\infty} x_n = 1$$

References